CHAPTER 4: LAWS OF MOTION

Advanced Physics Revision & Formula Masterclass | CBSE / JEE / NEET MUST KNOW EXAM FAVOURITE

1. MOMENTUM & NEWTON'S SECOND LAW

Linear Momentum & Newton's Second Law

FUNDAMENTAL

$$\vec{p} = m\vec{v}, \quad \vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} = m\vec{a}$$

Symbols: $\vec{p}$ = momentum, m = mass, $\vec{v}$ = velocity

SI Unit: $\text{kg}\cdot\text{m/s}$ or $\text{N}\cdot\text{s}$[cite: 31]

Component Form: $F_x = m a_x, \; F_y = m a_y, \; F_z = m a_z$[cite: 19]

Key Rule: Force at an instant determines acceleration at that exact instant[cite: 19].

Impulse-Momentum Theorem

SHORTCUT

$$\vec{J} = \int \vec{F}\, dt = \Delta\vec{p} = \vec{p}_{\text{final}} - \vec{p}_{\text{initial}}$$

Application: Impulsive forces acting for very short durations[cite: 20].

Rebound Wall Collision: Normal impulse for speed u at angle θ is $J = 2mu\cos\theta$[cite: 21].

2. NEWTON'S THIRD LAW & CONSERVATION OF MOMENTUM

Action-Reaction & Conservation Principle

MUST KNOW

$$\vec{F}_{AB} = -\vec{F}_{BA}, \quad \sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}} \text{ (for isolated system)}$$

Crucial Note: Action and reaction act on *different* bodies; they never cancel each other out[cite: 21].

Gun Recoil: $m_s v_s + m_g v_g = 0 \implies v_g = -\frac{m_s v_s}{m_g}$[cite: 21]

3. EQUILIBRIUM OF A PARTICLE

Concurrent Forces in Equilibrium

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$$\sum \vec{F} = 0 \implies \sum F_x = 0,\ \sum F_y = 0,\ \sum F_z = 0$$

Lami's Theorem (3 Forces): $\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$

Triangle Law: Three forces in equilibrium form a closed triangle with cyclic arrows[cite: 22].

CHAPTER 4: LAWS OF MOTION (CONTD.)

Friction, Circular Motion, and Mechanics Problem-Solving Methods

HIGH WEIGHTAGE

4. FRICTION LAWS & MECHANICS

Static & Kinetic Friction

$$f_s \leq \mu_s N, \quad f_k = \mu_k N$$

Angle of Repose (θ): $\mu_s = \tan\theta_{\text{max}}$[cite: 25]

Note: $\mu_k < \mu_s$. Static friction is self-adjusting[cite: 26, 32].

Elevator / Lift Apparent Weight

$$R = m(g \pm a)$$

Upward Accel ($+a$): $R = m(g+a)$ (Feels heavier)

Downward Accel ($-a$): $R = m(g-a)$ (Feels lighter)

5. CIRCULAR MOTION & BANKING OF ROADS

Vehicle on Banked Road with Friction

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$$v_{\max} = \sqrt{Rg\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}, \quad v_0 = \sqrt{Rg\tan\theta}\ (\text{Optimum Speed})$$

Level Road Max Speed: $v_{\text{max}} = \sqrt{\mu_s Rg}$[cite: 27]

Centripetal Force: $F_c = \frac{mv^2}{R}$ directed towards the center[cite: 27].

6. SYSTEMATIC PROBLEM-SOLVING METHOD (FBD TECHNIQUE)

Standard 5-Step FBD Solution Framework:

1. **System Isolation:** Choose a convenient part of the assembly as your system[cite: 29].
2. **Free-Body Diagram (FBD):** Isolate the system and draw all external contact and field forces acting on it[cite: 30].
3. **Coordinate Axes:** Set orthogonal axes (e.g., parallel and perpendicular to an incline).
4. **Apply Equations of Motion:** Use $\sum F_x = m a_x$ and $\sum F_y = m a_y$ (or $\sum F = 0$ for equilibrium).
5. **Newton's Third Law pairing:** Link internal forces across connected bodies (e.g., strings T, normal contacts N)[cite: 30].

7. SUMMARY TABLE OF CONSTANTS & UNITS

Physical Quantity	Symbol	SI Unit	Dimensional Formula